

PETER CAISSE

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Senior engineer with 15 years of experience building software. Skilled at technical leadership, project ownership, and cross-team collaboration. Loves to learn and share knowledge with others.

Technical Skills

- JavaScript/TypeScript
- Python
- Rust
- Postgres & PostGIS
- Linux (Debian/Ubuntu)
- Docker/Docker Compose
- AWS
- Bash
- Kubernetes
- Terraform

Professional Experience

Umbra, Santa Barbara, CA

03/2023 - Present

Senior Software Engineer II (Order and Delivery Team)

- Championed transitioning image delivery from Celery (Python library) to Argo Workflows (Kubernetes), dramatically improving observability and resiliency
- Implemented delivery pipeline enhancements to support transferring large multistatic files for DARPA's DRIFT program by parallelizing file copying and introducing resumption for timeouts and transient errors during multipart upload
- Systematically drove down delivery tech debt, sharply reducing delivery production incidents (compare ~400 collects whose delivery was delayed due to a production incident in 2024 vs. <10 in 2025)
- Improved observability via new collect statuses to better reflect where a collect is in the workflow internally and for customers and enhanced alerting
- Provided quick and attentive support during production incidents with frequent communication to stakeholders and accompanying comprehensive incident reports after the fact
- Implemented new capabilities to support extended dwell, extended scene sizes, and new product constraints (resolution/looks/scene size)
- Genericized and extended the existing delivery pipeline to support writing to GCS buckets in addition to S3 as required for high value contract
- Contributed extensively to UI and API development to support new scan imaging mode
- Made important contributions to the archive resell feature, including requirements definition, architecture, implementation, and optimization
- Spearheaded the effort to manage CVEs for Order and Delivery services for NIST compliance by developing and maintaining a framework for issue triage and a process surrounding it
- Made core internal Python library improvements that benefit all ground software teams
- Shaped how we code as a team, presenting those ideas at a Software Biweekly,

- making sure there was buy-in, and reinforcing them during code review
- Presented on static analysis tools (Pyright) to drive adoption on the team

Pandemic Response Lab (Opentrons), New York, NY 01/2022 - 01/2023
Engineering Lead 08/2022 - 01/2023

- Reorganized software department to form teams around projects
- Met with reports to hear concerns, understand team dynamics, give feedback, and discuss performance
- Developed HL7 templating system for high-impact, urgent state reporting initiative

Senior Software Engineer 01/2022 - 08/2022

- Quickly acquired key domain knowledge and deep understanding of code base (TypeScript to AWS Lambda, Postgres, Terraform)
- Ran and troubleshoot deployments and improved deployment process
- Fundamentally reworked specific workflow steps to support new PCR tests
- Strategically addressed tech debt, focusing on highest impact DX issues to dramatically increase the team's productivity
- Organized bi-weekly tech talk to exchange knowledge and build engineering culture

Azavea, Philadelphia, PA

Software Engineer 12/2017 - 01/2022

- Led greenfield project to develop a histopathology app for the client with follow up contracts and total revenue of over \$1 million
 - Met weekly with client to demo features and build roadmap
 - Worked with domain experts to understand technical and regulatory requirements
 - Implemented key features for frontend (TypeScript), API (Scala), and image processing (Python)
 - Led production support effort to triage, mitigate, and fix issues
- Made key contributions to open-source redistricting app called [DistrictBuilder](#)
- Mentored junior engineers
- Introduced a frontend stack of React/Redux/TypeScript which became standard on team and was used for all new projects
- Gave internal technical talks and wrote high-traffic blog posts

PromptWorks, Philadelphia, PA

Software Engineer 03/2017 - 12/2017

- Developed mobile colorimetry app (React Native) with Bluetooth device integration via SDK backed by a Python admin
- Created take-home coding challenge for apprentice candidates
- Prepared estimations for prospective clients
- Led agile ceremonies and client demos
- Presented technical topics and published blog posts

Monetate, Conshohocken, PA
Software Engineer

08/2013 - 01/2017
01/2015 - 01/2017

- Rewrote client-facing admin in React with new API endpoints (Django)
- Built real-time analytics dashboard powered by Druid
- Developed common frontend components (React) and improved testing framework
- Gave technical presentations to peers on topics such as Django ORM and Testing in React
- Built upon infrastructure as code via Vagrant, Ansible, Packer, and CloudFormation to create new services

Javascript Engineer

08/2013 - 01/2015

- Built custom Javascript actions to run directly on client sites
- Emphasis on small file size (minimal dependencies), high performance, and coding defensively

MEI, Jenkintown, PA – Software Engineer, 2012

Ci&T, Campinas, Brazil – Web Developer & Tech Support, 2008

OCTO Technology, Paris, France – Software Developer, 2007

Drexel University, Philadelphia, PA – Web Programmer, 2006

Innovative Development, LLC, Doylestown, PA – IT Intern, 2005

Education

Universidade Estadual de Campinas (UNICAMP) - Campinas, Brazil
MA, Linguistics, 2012

Drexel University - Philadelphia, PA
BS, Information Systems, Minor in Business and French, 2009

Certifications

- [Algorithms on Graphs](#) (Nov 2017)
- [Algorithmic Toolbox](#) (Oct 2017)
- [Data Structures](#) (Mar 2017)
- [Divide and Conquer, Sorting and Searching, and Randomized Algorithms](#) (Feb 2017)

Code

- <https://github.com/pcaisse>
- <https://codeberg.org/pcaisse>

Languages - Native English, Fluent in Portuguese and French